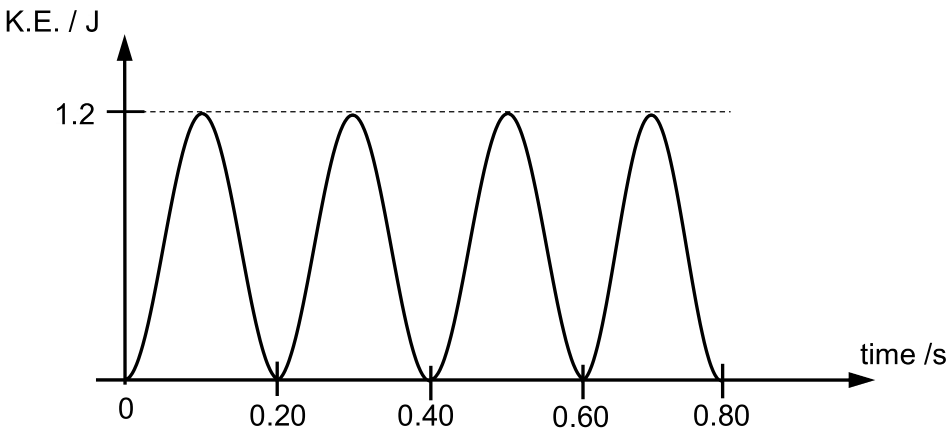
13 The graph below shows how the kinetic energy of a 50 g mass in simple harmonic motion varies with time.



What is the amplitude of the oscillation?

A 0.22 m **B** 0.44 m **C** 1.7 m **D** 2.8 m

Ans: B

From graph, KE_{max} = 1.2 J; T = 0.40 s

$KE_{max} = \frac{1}{2}\, m\, (\omega x_0)^2$

$1.2 = \frac{1}{2}\,(0.050)(2\pi/0.40)^2\, x_0^2$

x_0 = 0.44 m